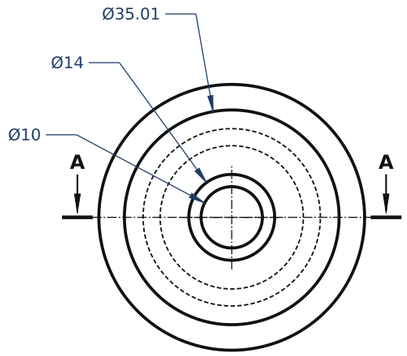
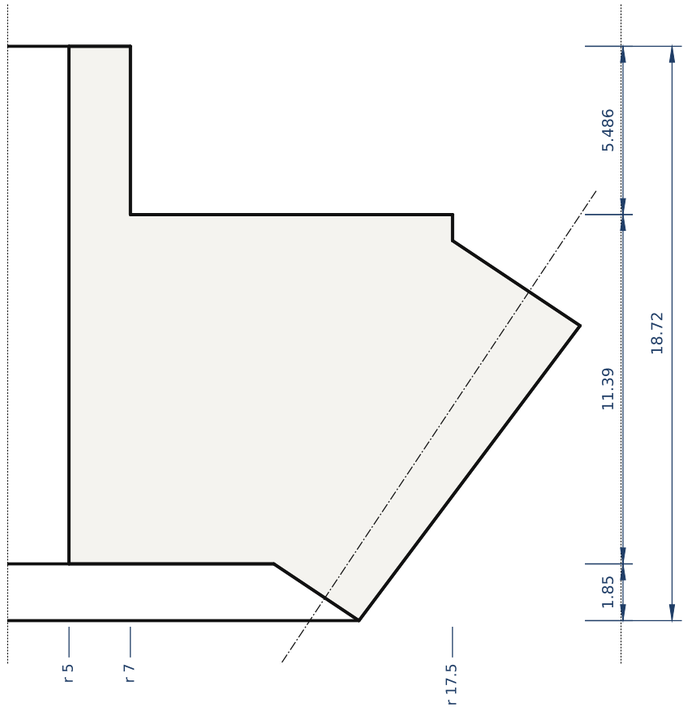
PLAN 1:1



SECTION A–A 5:1

(its part from r 3 to 23)



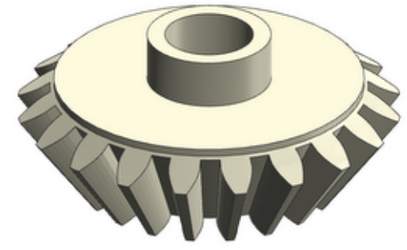
NOTES

1 POM-C, z 20, m 2, b 12: an idler, its two meshes balance, so it carries no torque on its shaft; pressed on its stainless stub shaft (DRV-104).
2 Its Ø14 hub bears on the inner 6000-2Z's inner ring; mounting distance 38.00 from the apex (§8).
3 Teeth cut (hob or shaper) or moulded; POM-C ≤ 60 °C (the VDI 2736-class limits of §5): bending 12 MPa, contact 30 MPa at the driver's limit (S 3.6 / 2.0).
4 The teeth as the 3-D view and the model draw them: the heel's involute profile on the back cone (Tredgold), the flanks straight to the apex, each tooth thinned 0.064 mm for the backlash (sim/unmodelled_parts.py); the section and the plan draw the blank, as a gear drawing does.
5 Status: the first cut of docs/drive-gear-belt.md on the record's 940 mm stack-up (z from the record's bottom end-bearing slot); the ledger asks for the drive to be re-cut on the layout (§6, item 71). PROPOSED.
6 General tolerances ISO 2768-mK; dimensions in mm; do not scale. Views by hidden-line removal of the model's solids.

REVISIONS

Rev	Date	Description
A	2026-10-11	First issue, draft for quotation

status	PROPOSED · the first cut, on the record's 940 mm stack-up: to re-cut on the layout (ledger §6, item 71)
qty per machine	3
used in	DRV-100
source	docs/drive-gear-belt.md §5
gear	straight bevel, 90° shaft angle; the teeth's proportions at the heel (h_a = m, h_f = 1.25 m)
teeth, module	z 20, m 2 at the heel; pressure angle 20°
pitch angle δ	33.69°; face 36.87°, root 29.72°
pitch Ø / tip Ø	40.00 / 43.33 at the heel
cone distance, face	R_e 36.06; b 12
mounting distance	38.00 ± 0.05, the cone's apex to the back face
mates with	DRV-101 and DRV-102, side gears z 30
backlash	j_n 0.10-0.15, set at assembly (docs/drive-gear-belt.md §5)
accuracy	ISO 17485 grade 8 class (AGMA Q 8, as sized) [IR]
mass each	0.0143 kg



3-D view (axonometric, not to scale)

DCCREG turbine · the drive: the reverser

Pinion, POM-C, z 20 m 2

Drawing no. **DCCREG-DRV-103**

Material: POM-C (acetal copolymer), natural, from annealed stock

Mass: 0.0143 kg	Qty per machine: 3	Rev A Sheet 75 of 103 A3
Scale: 1:1, 5:1	Tolerances: ISO 2768-mK	Date: 2026-10-11
Drawn: generated by docs/make_drawing_set.py	Checked: — Approved: —	**PROPOSED · DRAFT**